

Chapter 1

Standard Model of Cosmology

Contents

1.1	ΛCDM model	4
1.1.1	Overview - History of the Universe	5
1.1.2	Hubble Diagram	5
1.1.3	Big Bang Nucleosynthesis	5
1.1.4	Cosmic Microwave Background	5
1.1.5	Structure of the late Universe	5
1.1.6	Λ CDM model	5
1.2	The expanding Universe	5
1.2.1	Space-time metric	6
1.2.2	Geodesic equation	6
1.2.3	Cosmological distances	6
1.2.4	Hubble-Lemaitre law	6
1.3	The fundamental equations of Cosmology	6
1.3.1	General Relativity	6
1.3.2	Einstein equation	6
1.3.3	Friedmann equations	6
1.3.4	Boltzmann equations	6
1.4	History of the Universe	6
1.4.1	Big Bang Nucleosynthesis	6

1.4.2	Baryons Acoustic Oscillations	6
1.4.3	Recombination	6
1.4.4	Reionisation	6
1.4.5	Large Scale Structure	6
1.5	Problems of the Standard Model	6
1.5.1	Horizon Problem	6
1.5.2	Flatness Problem	6
1.5.3	Monopole Problem	6

Introduction

Cosmology can be described as the science of the Universe, dedicated to understanding its formation and evolution. Within the history of science and physics, cosmology holds a unique position: it is one of the few fields where we cannot replicate phenomena to directly test our theories. We have access to only one Universe, cannot reproduce its mechanisms, and cannot observe all of its information. This thesis aims to contribute to our understanding of the Universe's earliest moments. The first chapter provides a structured overview of the Standard Model of Cosmology, called Λ CDM, guiding the reader from its foundational concepts to its current limitations. The chapter is organized into five sections. The first introduces the essential elements of the model, while the second presents the fundamental tools and methods of cosmology. The third section develops the mathematical framework that underpins the model, followed by a discussion of the Universe's history in the fourth section. Finally, the fifth section highlights the open questions and challenges that continue to drive modern cosmological research.

ajouter que je me suis beaucoup inspiré de Dodelson-Schmidt ?

1.1 Λ CDM model

Pas sûr sur ce chapitre... Je voulais faire une sorte d'introduction vulgarisée mais c'est relativement difficile + peut-être pas utile...

1.1.1 Overview - History of the Universe

1.1.2 Hubble Diagram

1.1.3 Big Bang Nucleosynthesis

1.1.4 Cosmic Microwave Background

1.1.5 Structure of the late Universe

1.1.6 Λ CDM model

1.2 The expanding Universe

The fundamental milestone of Modern Cosmology was the discovery of its expansion, observed independently by Lemaître in 1927 [1] and Hubble in 1929 [2]. These results changed fundamentally the understanding of the Universe at that time, which was seen as immovable and eternal. This section focuses on the mathematical description of this expansion, and its implication on the Universe.

1.3 Expansion

The expansion of the Universe was proven by observing the velocity of galaxies around us. Lemaître and Hubble measured that the farther the galaxies were from us, the faster they moved away, involving an acceleration of the expansion of the universe. This expansion is characterise by the *scale factor* a .

- 1.3.1 Space-time metric
- 1.3.2 Geodesic equation
- 1.3.3 Cosmological distances
- 1.3.4 Hubble-Lemaitre law
- 1.4 The fundamental equations of Cosmology
 - 1.4.1 General Relativity
 - 1.4.2 Einstein equation
 - 1.4.3 Friedmann equations
 - 1.4.4 Boltzmann equations
- 1.5 History of the Universe
 - 1.5.1 Big Bang Nucleosynthesis
 - 1.5.2 Baryons Acoustic Oscillations
 - 1.5.3 Recombination
 - 1.5.4 Reionisation
 - 1.5.5 Large Scale Structure
- 1.6 Problems of the Standard Model
 - 1.6.1 Horizon Problem
 - 1.6.2 Flatness Problem
 - 1.6.3 Monopole Problem

Bibliography

- [1] G. Lemaître. “Un Univers homogène de masse constante et de rayon croissant rendant compte de la vitesse radiale des nébuleuses extra-galactiques”. In: *Annales de la Société Scientifique de Bruxelles* 47 (Jan. 1927), pp. 49–59.
- [2] Edwin Hubble. “A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae”. In: *Proceedings of the National Academy of Science* 15.3 (Mar. 1929), pp. 168–173. DOI: 10 . 1073 / pnas . 15 . 3 . 168.